

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

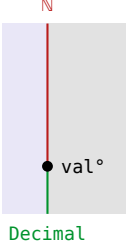
§6.1. Digits of a number

(6.1a)

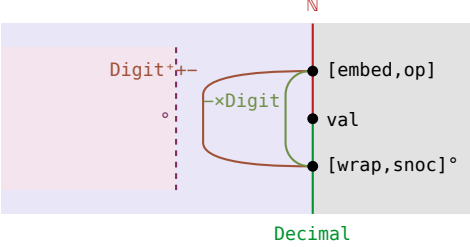
$\text{val}^\circ = \text{embed}^\circ \text{ wrap } \cup \text{ op}^\circ \text{ } \neg\text{Digit}(\text{val}^\circ) \text{ snoc}$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

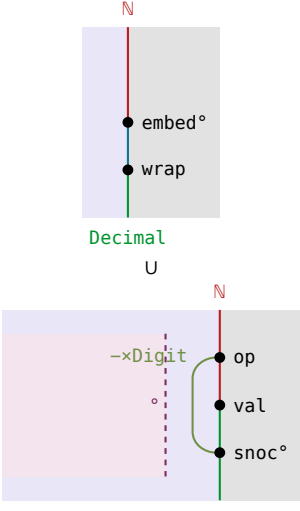
the converse of val , which reads digits as a number



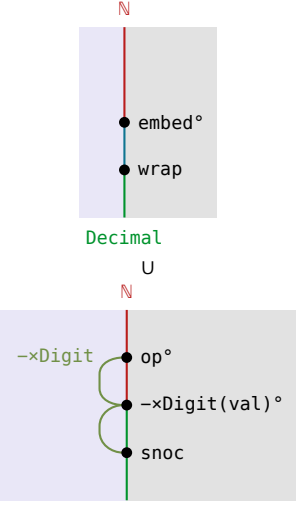
(a)



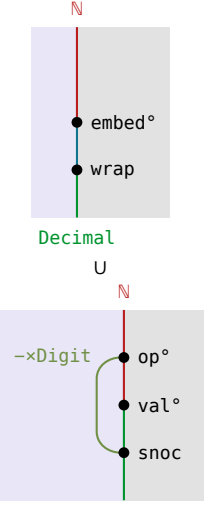
(b)



(c)



(d)



(e)

(a)	val°	val undoes wrap or snoc , reads the front with val and combines by embed or op , so its converse runs that backwards
(b)	$([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+--} (\neg\text{Digit}(\text{val}))) [\text{embed}, \text{op}]^\circ$	undoing wrap or snoc first gives two cases: a decimal and its last digit read by val then op , or one digit read by embed
(c)	$(\text{wrap}^\circ \text{ embed})^\circ \cup (\text{snoc}^\circ \neg\text{Digit}(\text{val}) \text{ op})^\circ$	the converse of each case is its steps undone in reverse: undo op , undo reading the front, then snoc ; or undo embed , then wrap
(d)	$\text{embed}^\circ \text{ wrap } \cup \text{ op}^\circ \neg\text{Digit}(\text{val})^\circ \text{ snoc}$	undoing the read of the front while keeping the last digit is val° on the front alone
(e)	$\text{embed}^\circ \text{ wrap } \cup \text{ op}^\circ \neg\text{Digit}(\text{val}^\circ) \text{ snoc}$	

(6.1b)

$\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$

op° splits a number into its quotient and remainder by 10

$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$

op° gives a pair with a nonzero first component exactly at the numbers with two or more digits

$(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$

embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

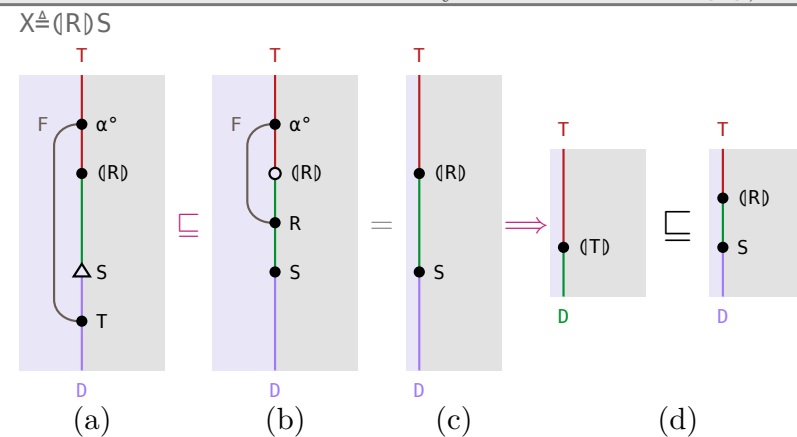
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

$$F(S) T \sqsubseteq R S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S

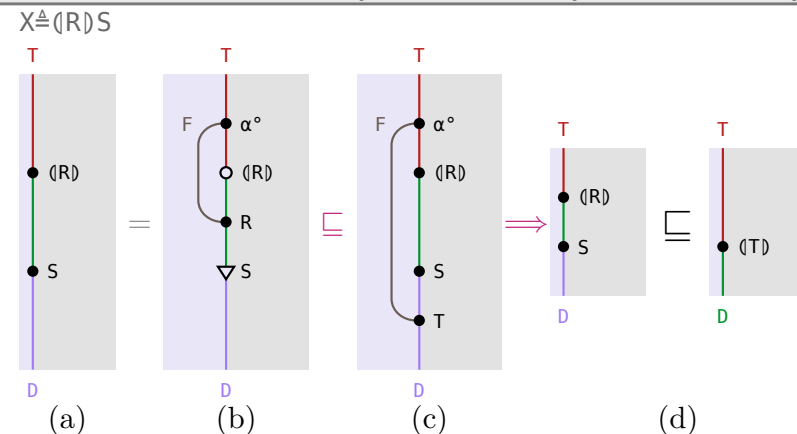


(a)	$\alpha^\circ F(\langle R \rangle) F(S) T$	$F(S) T \sqsubseteq R S$
(b)	$\alpha^\circ F(\langle R \rangle) R S$	$F(\langle R \rangle) R = \alpha(\langle R \rangle)$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce
(c)	$\langle R \rangle S$	$\langle R \rangle S$ is a prefix point of $\alpha^\circ F(-) T$, and $\langle T \rangle$ is the least one — (6.2) $\alpha^\circ F(X) T \sqsubseteq X \Rightarrow \langle T \rangle \sqsubseteq X$
(d)	$\alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$	

$$R S \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



(a)	$\langle R \rangle S$	$\alpha(\langle R \rangle) = F(\langle R \rangle) R$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce
(b)	$\alpha^\circ F(\langle R \rangle) R S$	$R S \sqsubseteq F(S) T$
(c)	$\alpha^\circ F(\langle R \rangle S) T$	$\langle R \rangle S$ is a postfix point of $\alpha^\circ F(-) T$, and $\langle T \rangle$ is the greatest one — (6.3) $X \sqsubseteq \alpha^\circ F(X) T \Rightarrow X \sqsubseteq \langle T \rangle$
(d)	$\langle R \rangle S \sqsubseteq \alpha^\circ F(\langle R \rangle S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$	

§6.3. Hylomorphisms

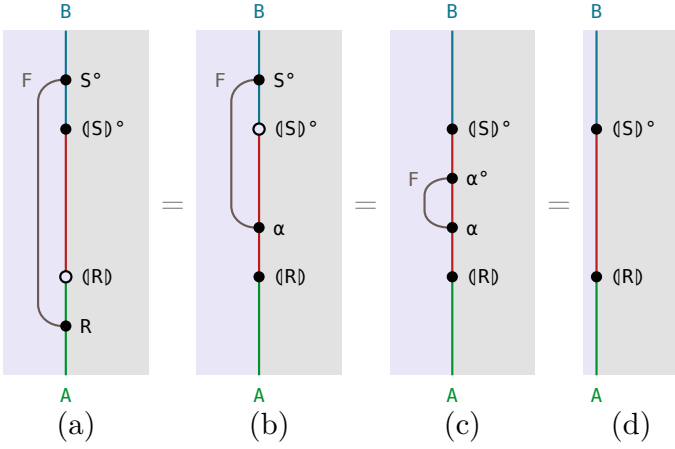
$$(\mathbb{S})^\circ (\mathbb{R}) = (\mu X : S^\circ F(X) R) \quad (6.3a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R = (\mathbb{S})^\circ (\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ (\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial

the body at $(\mathbb{S})^\circ (\mathbb{R})$

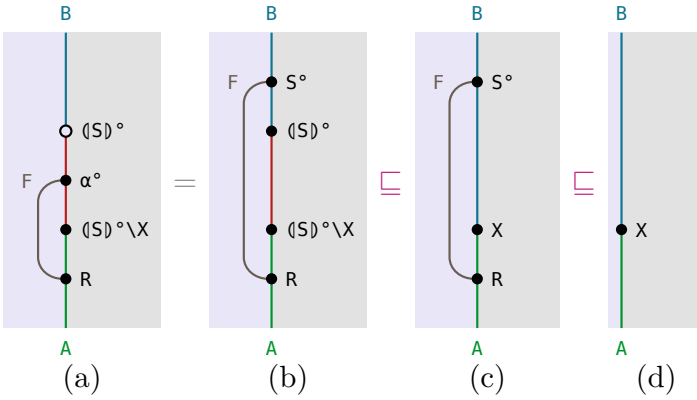


(a)	$S^\circ F((\mathbb{S})^\circ) F((\mathbb{R})) R$	$F((\mathbb{R})) R = \alpha(\mathbb{R}) \triangleq \text{reduce}$
(b)	$S^\circ F((\mathbb{S})^\circ) \alpha (\mathbb{R})$	$(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$
(c)	$(\mathbb{S})^\circ \alpha^\circ \alpha (\mathbb{R})$	$\alpha \text{ iso}$
(d)	$(\mathbb{S})^\circ (\mathbb{R})$	

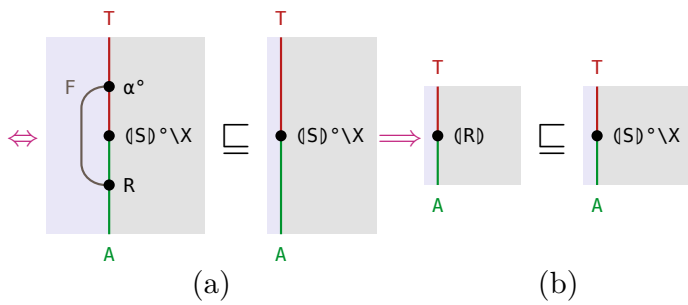
$$S^\circ F(X) R \sqsubseteq X \implies (\mathbb{S})^\circ (\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ (\mathbb{R})$ is included in X if X satisfies the associated recursion inequation

$$Y := (\mathbb{S})^\circ \setminus X$$



(a)	$(\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R$	$(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ)$
(b)	$S^\circ F((\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X)) R$	$(\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \text{ — adj-all}$
(c)	$S^\circ F(X) R$	$S^\circ F(X) R \sqsubseteq X$
(d)	X	$S \cdot \neg S \text{ — adj-all}$

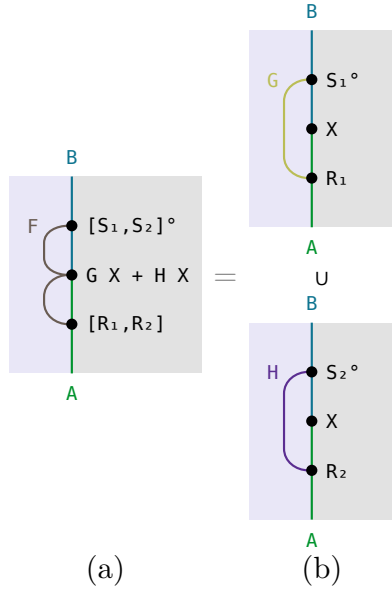


(a)	$S^\circ F(X) R \sqsubseteq X \implies \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \sqsubseteq (\mathbb{S})^\circ \setminus X$	$(\mathbb{R}) = (\mu X : \alpha^\circ F(X) R) \triangleq \text{reduce, (6.2b), adj-all}$
(b)	$S^\circ F(X) R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ \setminus X$	

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \Rightarrow ([S_1, S_2])^\circ ([R_1, R_2]) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

the body of Theorem 6.2 at $F(X)=G(X)+H(X)$

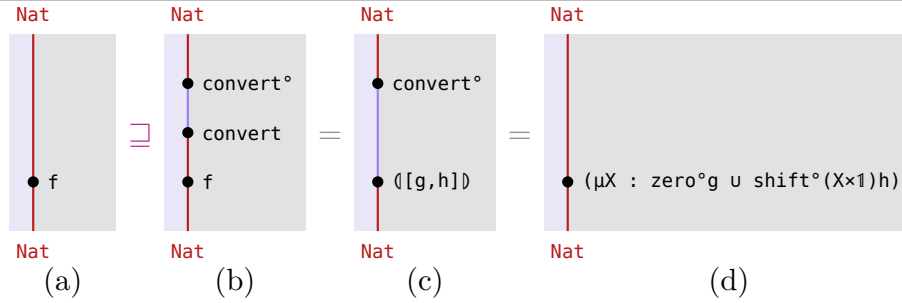


(a)	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$[S, T] = (S+T)\nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the + stands for the codiagonals ∇°, ∇ , which are not drawn
(b)	$S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$	$\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once f sends **zero** to g and turns **shift** into h , the recursion that halves the argument at each step computes f



(a)	f	convert simple
(b)	$\text{convert}^\circ \text{convert } f$	fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$
(c)	$\text{convert}^\circ ([g, h])$	Corollary 6.1, (6.3b)
(d)	$(\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h)$	

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes a to the power b

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves a at each step computes the remainder of a by b

$\text{zero } \text{mod}(b) = \text{zero}$

$0 \text{ mod } b$ is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

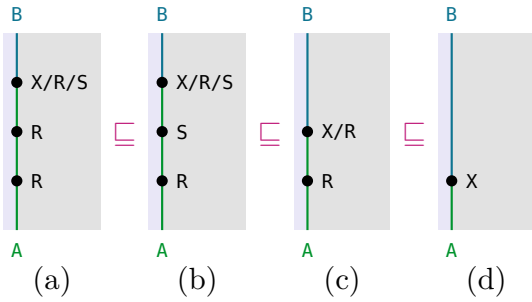
$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \ R \sqsubseteq S \ R \implies \text{inductive}(R)$$

(6.5a)

R is inductive when $RR \sqsubseteq SR$ for an inductive S



(a)	$(X / R / S) R R$	$RR \sqsubseteq SR$
(b)	$(X / R / S) S R$	division by S
(c)	$(X / R) R$	division by R
(d)	X	

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of $F \times G$ is a member of either component

$$\text{inductive}(S \text{ member}) \wedge X = S F(X) R \wedge Y = S F(Y) R \implies X = Y$$

(6.5c)

when $S \text{ member}$ is inductive, two solutions X, Y of $X = SF(X)R$ are equal

$$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S F(X) R \sqsubseteq X \implies \text{Entire}(X)$$

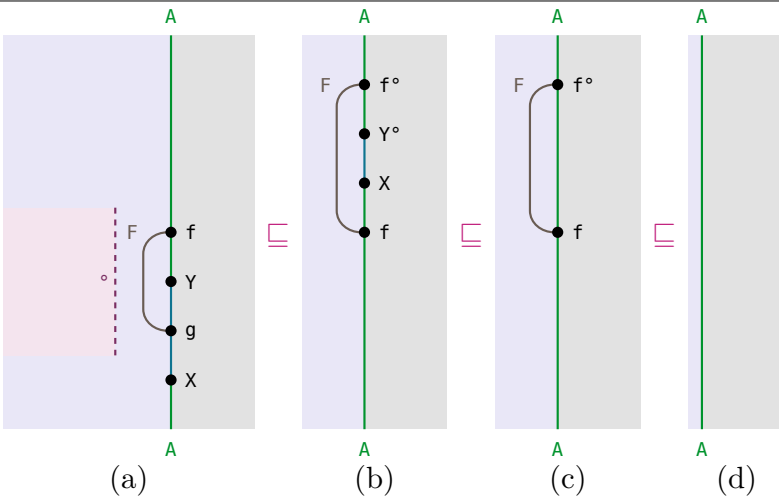
(6.5d)

when $S \text{ member}$ is inductive and S, R are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire

$$\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = g F(X) f \implies \text{Map}(X)$$

(6.5e)

when $g \text{ member}$ is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

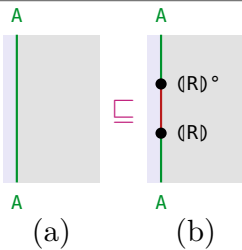


(a)	$(g F(Y) f)^\circ X$	$X = gF(X)f, g \text{ simple}$
(b)	$f^\circ F(Y^\circ X) f$	$Y^\circ X \sqsubseteq 1$
(c)	$f^\circ f$	$f \text{ simple}$
(d)	1	

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \implies 1 \sqsubseteq (R)^\circ (R)$$

(6.5f)

when $R^\circ \text{ member}$ is inductive and R is surjective, the fold (R) is surjective

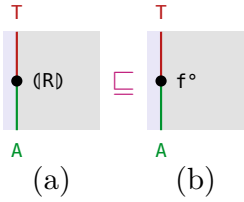


(a)	1	(6.5d) at $S = R^\circ$
(b)	$(R)^\circ (R)$	

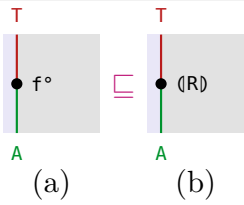
$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \implies f^\circ = \langle R \rangle$$

(6.5g)

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold $\langle R \rangle$ as its converse



(a)	$\langle R \rangle$	shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$
(b)	f°	

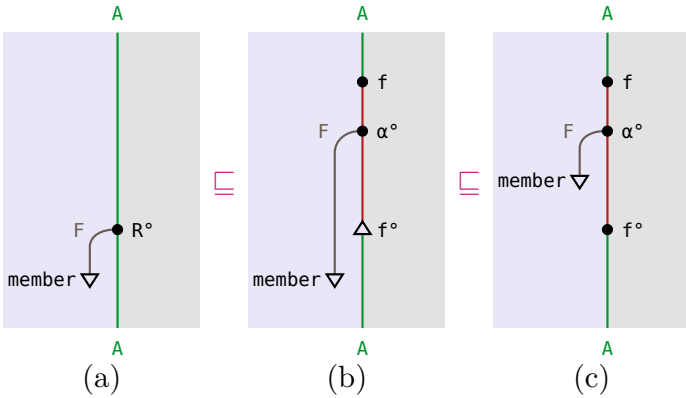


(a)	f°	$(6.5f)$ and $\langle R \rangle f \sqsubseteq 1$
(b)	$\langle R \rangle$	

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \implies R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ$$

(6.5h)

$R^\circ \text{ member}$ is below $\alpha^\circ \text{ member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{ member}$ is



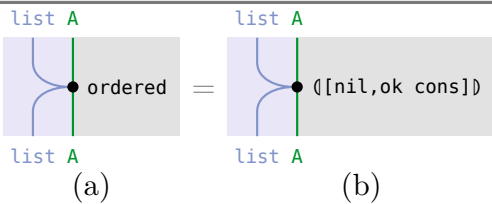
(a)	$R^\circ \text{ member}$	$R^\circ \sqsubseteq f \alpha^\circ F(f^\circ) \text{ — } R f \sqsubseteq F(f) \alpha$, shunting, converse
(b)	$f \alpha^\circ F(f^\circ) \text{ member}$	member lax natural
(c)	$f \alpha^\circ \text{ member } f^\circ$	

§6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

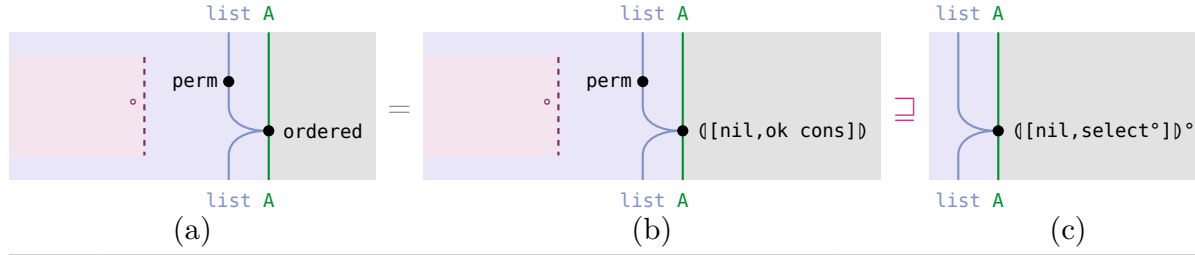


(a)	ordered	fold uniqueness
(b)	$\langle [\text{nil}, \text{ok cons}] \rangle$	

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered} \quad (6.6b)$$

every output of unfolding the input by **select** is a sorted permutation of it

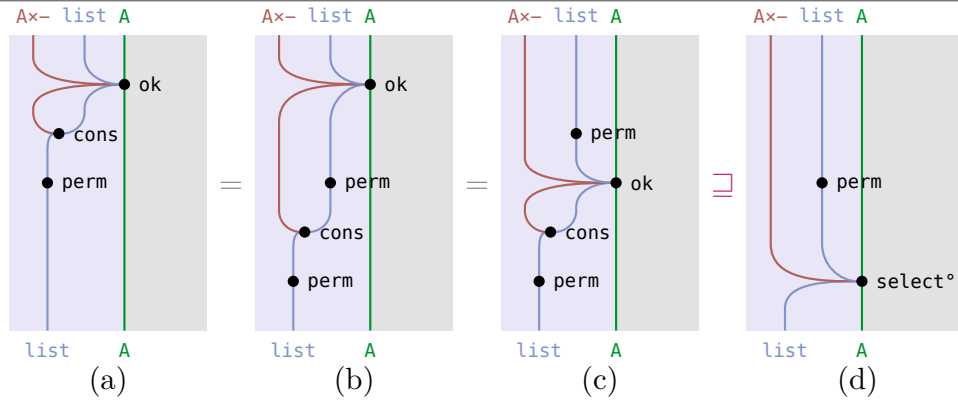
$\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$



(a)	$(\text{ordered perm})^\circ$	(6.6a)
(b)	$(([\text{nil}, \text{ok cons}])^\circ \text{ perm})^\circ$	fusion (6.4) under (6.6c)
(c)	$([\text{nil}, \text{select}^\circ])^\circ$	

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{ select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

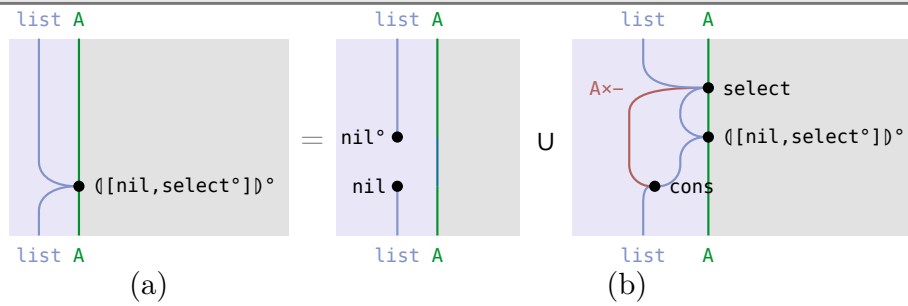
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



(a)	ok cons perm	$\text{cons perm} = (1 \times \text{perm}) \text{ cons perm}$
(b)	$\text{ok } (1 \times \text{perm}) \text{ cons perm}$	Ex 6.22, $\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ ok}$
(c)	$(1 \times \text{perm}) \text{ ok cons perm}$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
(d)	$(1 \times \text{perm}) \text{ select}^\circ$	

$$([\text{nil}, \text{select}^\circ])^\circ = \text{nil}^\circ \text{ nil } \cup \text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons} \quad (6.6d)$$

selection sort returns [] on [], and otherwise selects (a,y), sorts y and conses a back on

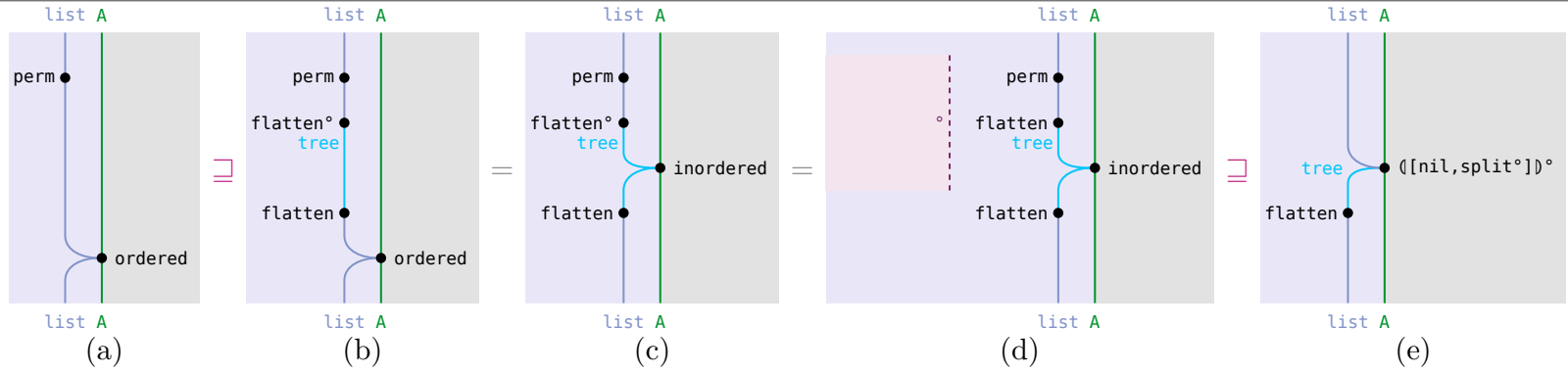


(a)	$([\text{nil}, \text{select}^\circ])^\circ$	unfolding the converse of a fold
(b)	$\text{nil}^\circ \text{ nil } \cup \text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons}$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \quad (6.6e)$$

flatten $\sqsubseteq$ **perm ordered**

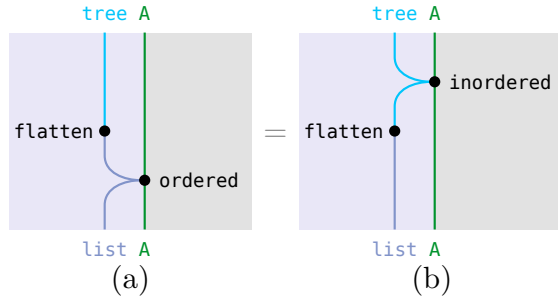
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



(a)	perm ordered	flatten is simple
(b)	perm flatten° flatten ordered	flatten ordered=inordered flatten
(c)	perm flatten° inordered flatten	converses
(d)	(inordered flatten perm)° flatten	fusion (6.4) under (6.6j)
(e)	$([\text{nil}, \text{split}^\circ])^\circ$ flatten	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

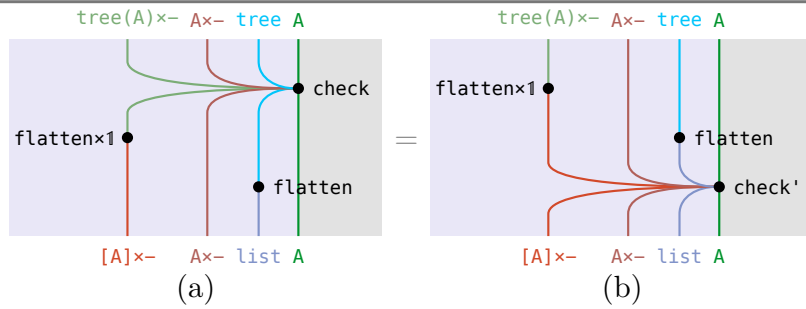
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



(a)	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
(b)	inordered flatten	

$$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{check}' \quad (6.6g)$$

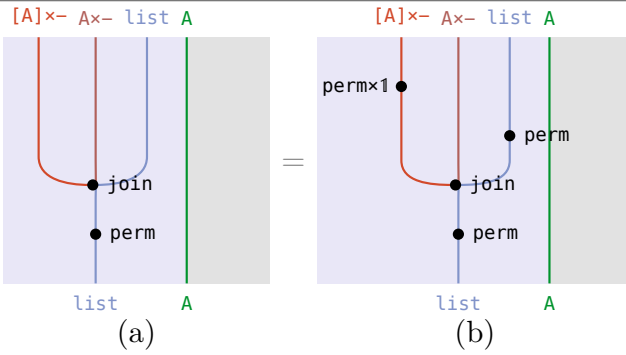
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



(a)	check (flatten $\times$ (1 $\times$ flatten))	b intree x $\iff$ b inlist flatten(x)
(b)	(flatten $\times$ (1 $\times$ flatten)) check'	

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm} \quad (6.6h)$$

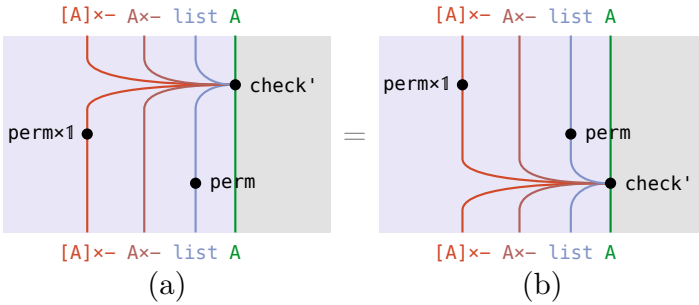
permuting the list $x \mathrel{++} [a] \mathrel{++} y$ is the same as first permuting x and y and then permuting the joined list



(a)	join perm	perm is transitive and closed under ++
(b)	(perm × (1 × perm)) join perm	

$$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{check}' \quad (6.6i)$$

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

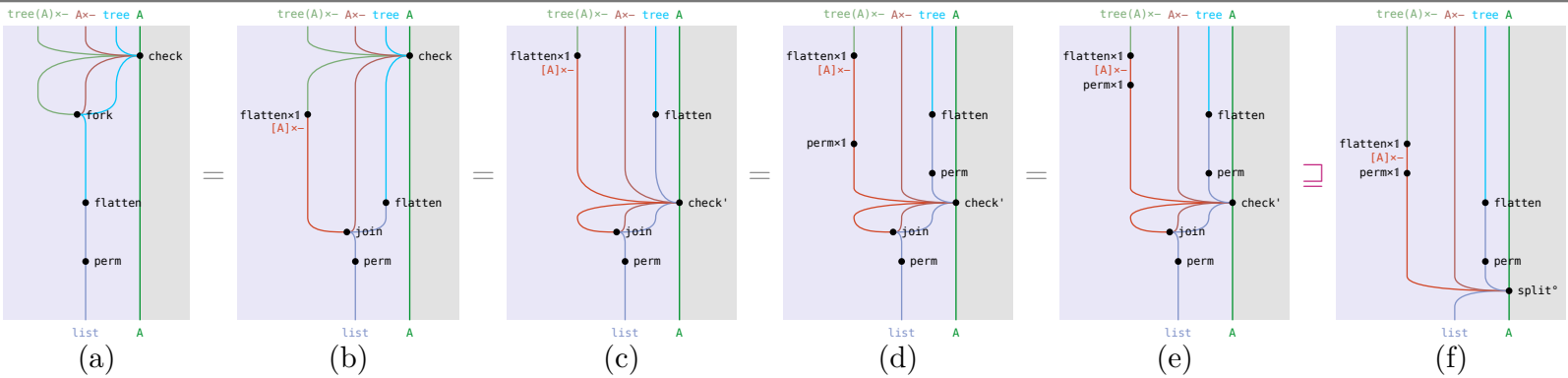


(a)	check' (perm × (1 × perm))	$b \text{ inlist } x \iff b \text{ inlist perm}(x)$
(b)	(perm × (1 × perm)) check'	

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check fork} \quad (6.6j)$$

flatten perm

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



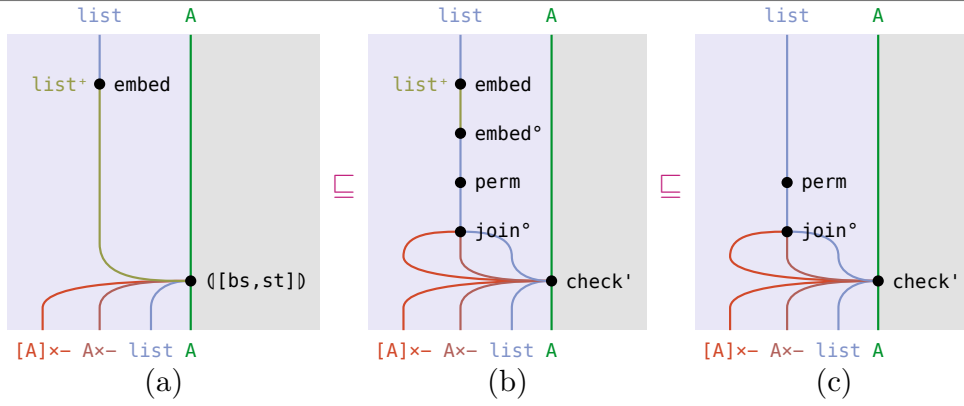
(a)	check fork flatten perm	fork flatten=F(flatten)join
(b)	check (flatten × (1 × flatten)) join perm	check F(flatten)=F(flatten)check'
(c)	(flatten × (1 × flatten)) check' join perm	join perm=F(perm)join perm, check' F(perm)=F(perm)check'
(d)	(flatten × (1 × flatten)) (perm × (1 × perm)) check' join perm	functors
(e)	((flatten perm) × (1 × (flatten perm))) check' join perm	split ^o ⊆ check' join perm
(f)	((flatten perm) × (1 × (flatten perm))) split ^o	

$\text{bs} \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed}$ (6.6k)

$\langle [\text{bs}, \text{st}] \rangle \sqsubseteq \text{perm join}^\circ \text{check}'$

turning a list into a non-empty list and folding it with any $[\text{bs}, \text{st}]$ meeting the two conditions below gives only

checked triples (x, a, y) whose join $x \# [a] \# y$ is a permutation of the list

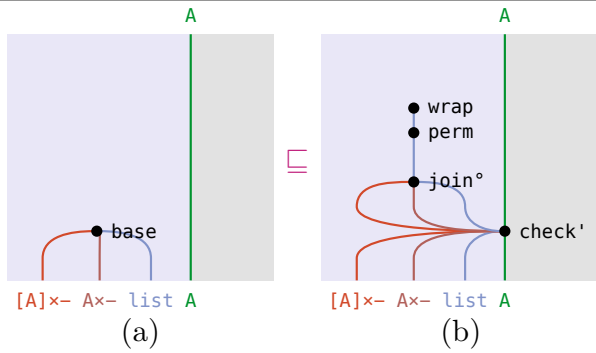


(a)	$\text{embed } \langle [\text{bs}, \text{st}] \rangle$	(6.6l) and (6.6m), induction on the non-empty list
(b)	$\text{embed embed}^\circ \text{perm join}^\circ \text{check}'$	embed is simple
(c)	$\text{perm join}^\circ \text{check}'$	

$\text{base} \sqsubseteq \text{wrap perm join}^\circ \text{check}'$

(6.6l)

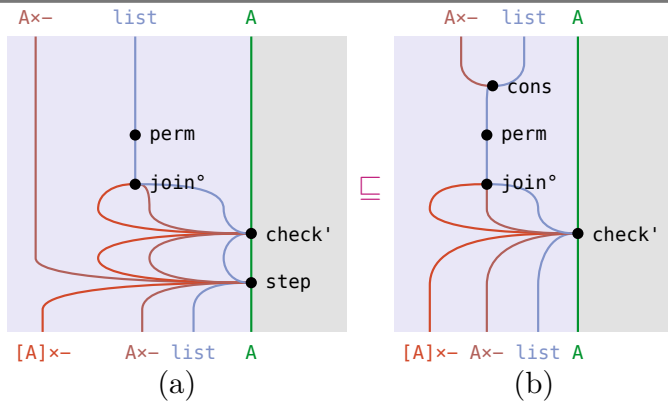
base(a) is a checked triple whose join is a permutation of the one-element list $[a]$



(a)	base	$[] \# [a] \# [] = [a]$, and check' holds on empty lists
(b)	$\text{wrap perm join}^\circ \text{check}'$	

$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ \text{check}')) \text{step} \sqsubseteq \text{cons perm join}^\circ \text{check}'$ (6.6m)

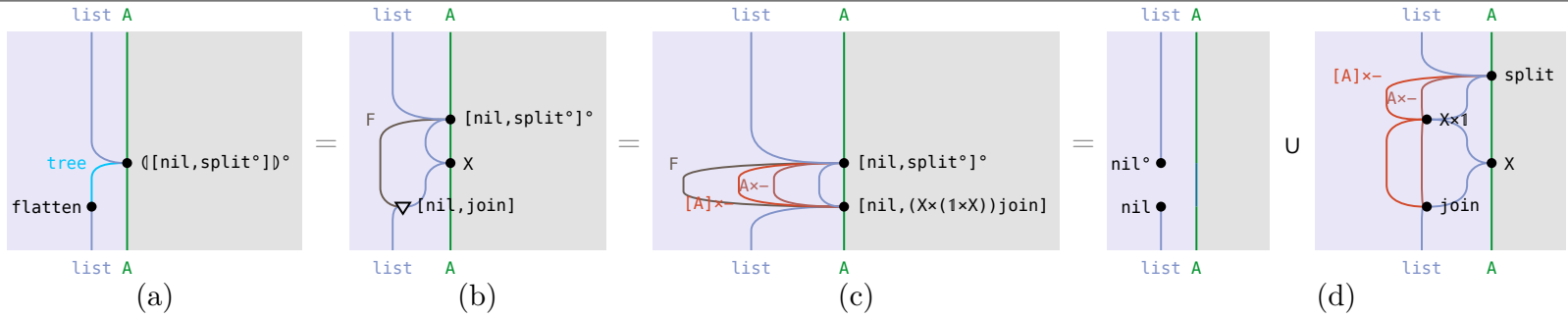
if (x, b, y) is a **checked triple** for a permutation of $\mathbf{l}$, then $\text{step}(a, (x, b, y))$ is a **checked triple** for a permutation of $[a] \# \mathbf{l}$



(a)	$(1 \times (\text{perm join}^\circ \text{check}')) \text{step}$	aRb puts a in front of x ; otherwise bRa and a goes in front of y
(b)	$\text{cons perm join}^\circ \text{check}'$	

$$([nil, split]^{\circ})^{\circ} \text{ flatten} = nil^{\circ} nil \cup split \left(([nil, split]^{\circ})^{\circ} \text{ flatten} \right) \times (1 \times ([nil, split]^{\circ})^{\circ} \text{ flatten})) \text{ join} \quad (6.6n)$$

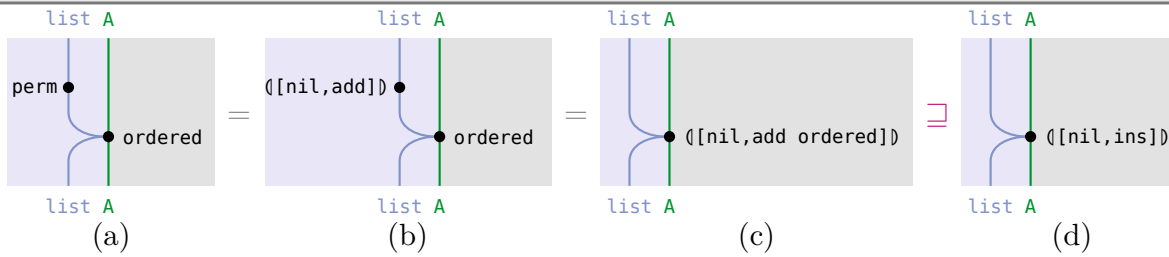
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



(a)	$([nil, split]^{\circ})^{\circ} \text{ flatten}$	hyломorphism theorem
(b)	$[nil, split]^{\circ} F(X) [nil, join]$	$F(X) [nil, join] = [nil, (X \times 1 \times X) join]$
(c)	$[nil, split]^{\circ} [nil, (X \times (1 \times X)) join]$	coproduct
(d)	$nil^{\circ} nil \cup split (X \times (1 \times X)) join$	

$$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow ([nil, ins]) \sqsubseteq \text{perm ordered} \quad (6.6o)$$

for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations

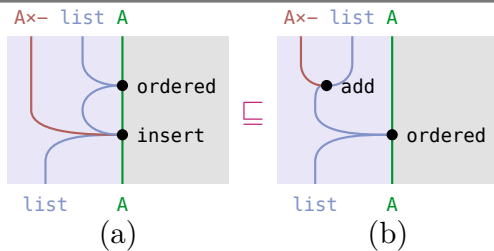


(a)	perm ordered	perm = $([nil, add])$
(b)	$([nil, add])$ ordered	fusion, add ordered = $(1 \times \text{ordered})$ add ordered
(c)	$([nil, add ordered])$	$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$
(d)	$([nil, ins])$	

$$(\forall a. \forall b. \text{leb}(a,b) = \text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b) = \text{false} \Rightarrow R(b,a)) \wedge (\forall a. \forall b. \forall c. \quad (6.6p)$$

$$R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{ insert} \sqsubseteq \text{add ordered}$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



(a)	$(1 \times \text{ordered}) \text{ insert}$	insert splices a in, and R transitive and total
(b)	add ordered	

§6.7. Closure

$$R^* = (\mu X : 1 \cup X R) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***

$$1 \sqsubseteq X \wedge X X \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R***

$$1 \sqsubseteq R^* \quad (6.7c)$$

R* is reflexive

	(a)	1	
$\sqsubseteq$	(b)	$1 \cup R R^*$	union
$=$	(c)	R^*	fixed point, Thm 6.1

$$\mathbf{R} \subseteq \mathbf{R}^*$$

(6.7d)

 $\mathbf{R}^*$ contains $\mathbf{R}$

	(a)	$\mathbf{R}$	
=	(b)	$\mathbf{R}$	identity
$\sqsubseteq$	(c)	$\mathbf{R} \mathbf{R}^*$	$\mathbf{1} \sqsubseteq \mathbf{R}^*$ — (6.7c)
$\sqsubseteq$	(d)	$\mathbf{R}^*$	fixed point

$$\mathbf{R}^* \mathbf{R}^* \subseteq \mathbf{R}^*$$

(6.7e)

 $\mathbf{R}^*$ is transitive, because $\mathbf{R}^*/\mathbf{R}^*$ is a prefixed point of $\mathbf{X} \mapsto \mathbf{1} \cup \mathbf{R} \mathbf{X}$

	(a)	$(\mathbf{1} \cup \mathbf{R} (\mathbf{R}^* / \mathbf{R}^*)) \mathbf{R}^*$	
=	(b)	$\mathbf{R}^* \cup \mathbf{R} (\mathbf{R}^* / \mathbf{R}^*) \mathbf{R}^*$	$(\mathbf{P} \cup \mathbf{Q}) \mathbf{T} = \mathbf{P} \mathbf{T} \cup \mathbf{Q} \mathbf{T}$ — composition distributes over $\cup$
$\sqsubseteq$	(c)	$\mathbf{R}^* \cup \mathbf{R} \mathbf{R}^*$	$(\mathbf{S} / \mathbf{S}) \mathbf{S} \subseteq \mathbf{S}$
$\sqsubseteq$	(d)	$\mathbf{R}^*$	$\mathbf{R} \mathbf{R}^* \subseteq \mathbf{R}^*$ — (6.7d)

$$\mathbf{1} \in \mathbf{X} \wedge \mathbf{X} \mathbf{X} \subseteq \mathbf{X} \wedge \mathbf{R} \subseteq \mathbf{X} \implies \mathbf{R}^* \subseteq \mathbf{X}$$

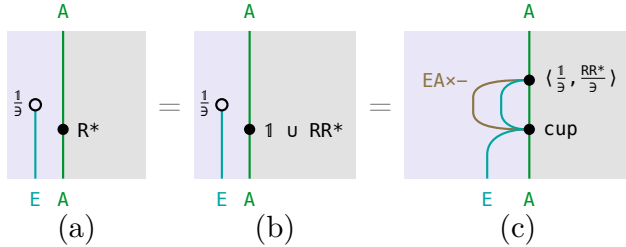
(6.7f)

a reflexive transitive $\mathbf{X}$ containing $\mathbf{R}$ is a prefixed point of $\mathbf{X} \mapsto \mathbf{1} \cup \mathbf{R} \mathbf{X}$, so it contains $\mathbf{R}^*$

	(a)	$\mathbf{1} \cup \mathbf{R} \mathbf{X}$	
$\sqsubseteq$	(b)	$\mathbf{X} \cup \mathbf{X} \mathbf{X}$	$\mathbf{1} \in \mathbf{X}, \mathbf{R} \subseteq \mathbf{X}$
$\sqsubseteq$	(c)	$\mathbf{X}$	$\mathbf{X} \mathbf{X} \subseteq \mathbf{X}$

$$\frac{\mathbf{R}^*}{\mathfrak{Z}} = \left(\frac{\mathbf{1}}{\mathfrak{Z}}, \frac{\mathbf{R} \mathbf{R}^*}{\mathfrak{Z}} \right) \text{ cup}$$

(6.7g)

the set of $\mathbf{R}^*$ -successors of $\mathbf{a}$ is $\mathbf{a}$ itself joined with the $\mathbf{R}^*$ -successors of its $\mathbf{R}$ -successors


(a)	$\frac{\mathbf{R}^*}{\mathfrak{Z}}$	$\mathbf{R}^* = \mathbf{1} \cup \mathbf{R} \mathbf{R}^*$
(b)	$\frac{\mathbf{1} \cup \mathbf{R} \mathbf{R}^*}{\mathfrak{Z}}$	$\wedge (\mathbf{R} \cup \mathbf{S}) = (\wedge (\mathbf{R}), \wedge (\mathbf{S})) \text{ cup}$
(c)	$\left(\frac{\mathbf{1}}{\mathfrak{Z}}, \frac{\mathbf{R} \mathbf{R}^*}{\mathfrak{Z}} \right) \text{ cup}$	

$$\mathbf{R} - \emptyset = \mathbf{R}$$

(6.7h)

taking nothing away leaves $\mathbf{R}$

$$\mathbf{R} \cup (\mathbf{S} - \mathbf{R}) = \mathbf{R} \cup \mathbf{S}$$

beside $\mathbf{R}$, only the part of $\mathbf{S}$ outside $\mathbf{R}$ adds anything

$$\mathbf{R} - (\mathbf{S} \cup \mathbf{T}) = \mathbf{R} - \mathbf{S} - \mathbf{T}$$

taking away $\mathbf{S}$ and then $\mathbf{T}$ takes away $\mathbf{S} \cup \mathbf{T}$

$$(\mathbf{R} \cup \mathbf{S}) - \mathbf{T} = (\mathbf{R} - \mathbf{T}) \cup (\mathbf{S} - \mathbf{T})$$

taking $\mathbf{T}$ away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu \mathbf{X} : \varphi(\psi(\mathbf{X}))) = \varphi(\mu \mathbf{Y} : \psi(\varphi(\mathbf{Y})))$$

(6.7i)

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

$$\mathbf{S} \mathbf{R}^* = (\mu \mathbf{X} : \mathbf{S} \cup \mathbf{X} \mathbf{R})$$

(6.7j)

 $\mathbf{S}$ followed by any number of $\mathbf{R}$ steps is the least $\mathbf{X}$ containing $\mathbf{S}$ and closed under a further $\mathbf{R}$ step

$$\mathbf{R}^* \mathbf{S} = (\mu \mathbf{X} : \mathbf{S} \cup \mathbf{R} \mathbf{X})$$

any number of $\mathbf{R}$ steps followed by $\mathbf{S}$ is the least $\mathbf{X}$ containing $\mathbf{S}$ and closed under an $\mathbf{R}$ step in front

$$\theta(\emptyset, \mathbf{S}) = \mathbf{S} \mathbf{R}^*$$

(6.7k)

started with nothing found and $\mathbf{S}$ to explore, θ computes $\mathbf{S} \mathbf{R}^*$

	(a)	$\theta(\emptyset, \mathbf{S})$	
=	(b)	$\emptyset \cup (\mu \mathbf{X} : \mathbf{S} \cup (\mathbf{X} \mathbf{R} - \emptyset))$	definition of θ
=	(c)	$(\mu \mathbf{X} : \mathbf{S} \cup \mathbf{X} \mathbf{R})$	$\mathbf{R} - \emptyset = \mathbf{R}$ — (6.7h)
=	(d)	$\mathbf{S} \mathbf{R}^*$	(6.7j)

$$\theta(P, \emptyset) = P$$

(6.71)

with nothing left to explore, θ returns what it has found

	(a)	$\theta(P, \emptyset)$	
=	(b)	$P \cup (\mu X : \emptyset \cup (X R - P))$	definition of θ
=	(c)	$P \cup \emptyset$	$\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$
=	(d)	P	$P \cup \emptyset = P$